

Supplementary Material: Within- and between-year seasonal patterns of respiratory pathogens and the potential for concurrent outbreaks

Keywords: respiratory pathogens, influenza, RSV, seasonal dynamics, survival analysis, concurrent outbreaks, geographic variation

1. Supplemental Materials

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1.1. Tables

1.1.1. Comparable Models by City and Pathogen

Table 1: Table S1: Number of comparable models ($\Delta AIC < 7$) per city and pathogen. This table illustrates the distribution of model availability across the eight study cities, highlighting the variation in the number of comparable models per city.

City	Influenza	RSV
<i>North</i>		
Beijing	20	39
Lanzhou	2	4
Xian	2	3
<i>North Total</i>	<i>24</i>	<i>46</i>
<i>South</i>		
Wuhan	20	22
Suzhou	18	3
Guangzhou	4	11
Wenzhou	2	2
Yunfu	2	2
<i>South Total</i>	<i>46</i>	<i>40</i>
Grand Total	70	86

1.1.2. Median Probability of Concurrent Outbreaks by Risk Threshold

Table 2: Table S2: Median probability of concurrent outbreaks by city and risk threshold. Values represent the median proportion of years with concurrent high-risk periods for both Influenza and RSV across 1000 simulation paths. Risk thresholds are defined as joint top 30%, 20%, and 10% percentiles of disease burden.

City	Top 30%	Top 20%	Top 10%
<i>North</i>			
Beijing	0.200	0.100	0.000
Lanzhou	0.118	0.059	0.000
Xian	0.450	0.350	0.150
<i>South</i>			
Wuhan	0.400	0.300	0.150
Suzhou	0.350	0.200	0.100
Guangzhou	0.188	0.125	0.062
Wenzhou	0.812	0.500	0.250
Yunfu	0.375	0.188	0.125

1.1.3. Median Time to Next Concurrent Outbreak by Risk Threshold

Table 3: Table S3: Median time to subsequent concurrent outbreak (years) by city and risk threshold, with 95% confidence intervals. Values represent the x-intercepts of the dashed vertical lines in Figure 6, indicating the time (in years) at which 50% of simulation paths are expected to experience their subsequent concurrent outbreak. Confidence intervals (in parentheses) reflect uncertainty in the Kaplan-Meier survival estimates, which properly account for right-censoring of paths that never experience a second outbreak. The relatively narrow confidence intervals reflect high precision from aggregating across 1000 simulation paths per city. Values marked as “—” indicate that the 50% threshold was not reached within the simulation window. Risk thresholds are defined as joint top 30%, 20%, and 10% percentiles of disease burden.

City	Top 30%	Top 20%	Top 10%
<i>North</i>			
Beijing	3.0 (2.0–3.0)	5.0 (4.0–6.0)	—
Lanzhou	5.0 (5.0–6.0)	—	—
Xian	1.0 (1.0–1.0)	2.0 (2.0–2.0)	3.0 (3.0–3.0)
<i>South</i>			
Wuhan	1.0 (1.0–1.0)	2.0 (2.0–2.0)	3.0 (3.0–3.0)
Suzhou	2.0 (2.0–2.0)	3.0 (3.0–3.0)	6.0 (5.0–6.0)
Guangzhou	3.0 (3.0–4.0)	6.0 (6.0–7.0)	—
Wenzhou	1.0 (1.0–1.0)	1.0 (1.0–1.0)	2.0 (2.0–2.0)
Yunfu	2.0 (2.0–2.0)	3.0 (3.0–3.0)	7.0 (6.0–8.0)

1.1.4. Optimal Within-Year and Between-Year Cycle Lengths

Table 4: Table S4: Optimal within-year and between-year cycle lengths (months) for each city and pathogen combination. Values represent the cycle lengths of the best-performing model (lowest AIC) for each city-pathogen pair. Within-year cycles are shown for both pathogens; between-year cycles are only shown for models that included multi-seasonal decomposition (MSTL). These values correspond to the data presented in Figure 3 (A) and (B) of the main text.

Region	City	Within-year cycle (months)		Between-year cycle (months)	
		Influenza	RSV	Influenza	RSV
North					
	Beijing	13.0	10.0	28.0	24.0
	Lanzhou	13.0	13.0	20.0	35.0
	Xian	13.0	13.0	36.0	33.0
South					
	Suzhou	14.0	12.0	26.0	21.0
	Wuhan	13.0	12.0	30.0	19.0
	Wenzhou	12.0	12.0	22.0	18.0
	Guangzhou	13.0	12.0	34.0	31.0
	Yunfu	14.0	10.0	21.0	24.0

1.2. Figures

1.2.1. Distribution of Between-Year Cycle Lengths

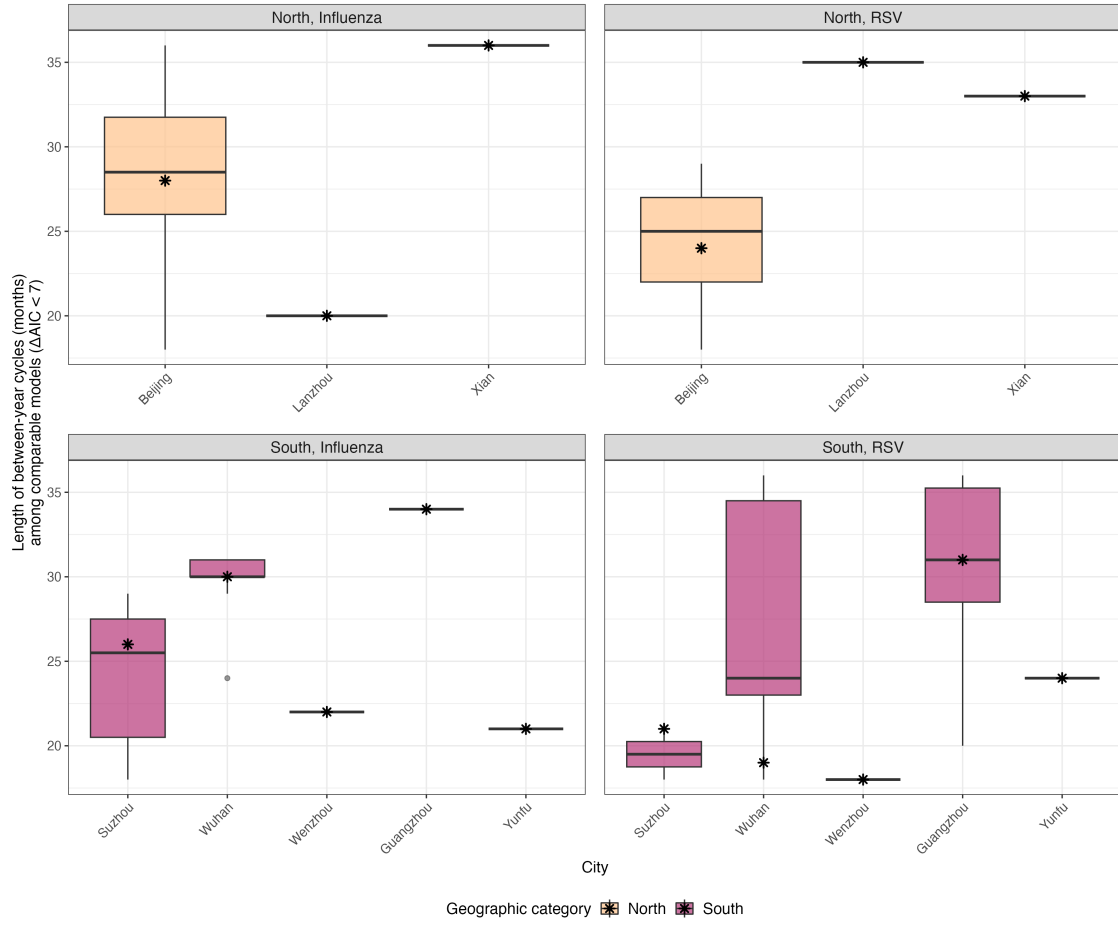


Figure 1: Figure S1: Distribution of between-year cycle lengths (months) among comparable models ($\Delta AIC < 7$) by city and pathogen, separated by geographic region (North vs South). Each panel displays boxplots of the between-year cycle lengths (in months) for each city within the specified region and pathogen combination. The asterisk (*) marks the best-performing model (lowest AIC) for each city-pathogen combination.

Supplementary Text 1. Literature search strategy

City-level surveillance time series for respiratory syncytial virus (RSV) and influenza virus (InfV) were assembled using a two-stage approach designed to support modelling of both within-year seasonality and between-year (interannual) variation. Consequently, we required year-indexed, continuous multi-year time series (e.g., 2017-01, 2017-02, ...), and excluded studies that only reported aggregated seasonal profiles obtained by averaging across years within each calendar month or epidemiological week (i.e., month-of-year or week-of-year values without a year index).

RSV datasets were first identified from an existing systematic review [2] and its accompanying extraction table, followed by rule-based filtering and manual verification of time-series structure. Influenza datasets were then assembled for the RSV-anchored study cities using a complementary strategy combining (i) targeted data-source mining from two recent influenza systematic reviews [1, 3] and (ii) city-specific PubMed searches. All PubMed searches were conducted on 12 August 2024 without language restrictions. Search strings and screening criteria are provided below.

RSV dataset identification

Candidate RSV studies were obtained from the extraction table released with the systematic review by Guo et al. [2] (`RSVseasonality_China_2019_foruploading.xlsx`; available at; 161 records). Records were first stratified by study setting, yielding 142 mainland China records after excluding Taiwan (10) and Hong Kong (9). Study periods were parsed for all mainland records (142/142) and filtered using pre-specified eligibility criteria: prefecture-/city-level surveillance (administrative level two), at least five consecutive years of observations (≥ 60 months), monthly or weekly temporal resolution, and laboratory-confirmed RSV indicators (e.g., RSV positivity among ARI/LRTI inpatients or outpatients). Applying these filters retained 20 mainland records with ≥ 5 years of observations and 19 records that were both ≥ 5 years and city-level.

Remaining records were then manually reviewed to verify that reported monthly/weekly values formed a continuous, year-indexed multi-year time series (e.g., 2017-01, 2017-02, ...), rather than an aggregated seasonal profile (a “climatological” seasonal cycle) computed by pooling across multiple years and summarizing by calendar month or epidemiological week (e.g., mean January value over 2017–2022; x-axis labelled Jan–Dec only). Six studies reporting only such multi-year month-of-year averages were excluded; as a consequence, Hangzhou and Shanghai no longer met the continuous time-series requirement. The final RSV dataset comprised eight cities (Beijing, Lanzhou, Xi’an, Wuhan, Suzhou, Wenzhou, Guangzhou, and

Yunfu). For each city, one study was selected and digitized to construct the RSV time series used in the analysis.

To cross-check coverage and to identify any additional eligible city-level RSV time series not captured by the review table, we also queried PubMed using the following string:

```
("Respiratory Syncytial Viruses"[Mesh] OR RSV[Title/Abstract]  
OR "respiratory syncytial"[Title/Abstract])  
AND (China[Mesh] OR China[Title/Abstract] OR Chinese[Title/Abstract])
```

Influenza dataset identification

Influenza time series were assembled for the same eight RSV-anchored cities (Beijing, Lanzhou, Xi'an, Wuhan, Suzhou, Wenzhou, Guangzhou, and Yunfu) using a two-pronged strategy.

First, we screened two influenza systematic reviews [1, 3] to identify (i) candidate primary studies reporting city-level influenza surveillance data for any of the eight target cities and (ii) descriptions of surveillance systems, case/virological definitions, and reporting frequency. Where a target city had an eligible influenza time series reported in these reviews (or in the primary studies referenced therein), the corresponding original articles were retrieved and assessed against the eligibility criteria described below.

Second, to minimize the risk of missing eligible city-level series not captured by the reviews, we performed city-specific PubMed searches for each target city using the following template (city term substituted as appropriate and including common romanization variants when applicable):

```
("Influenza, Human"[Mesh] OR influenza[Title/Abstract] OR ILI[Title/Abstract])  
AND (<CITY>[Title/Abstract])  
AND (China[Mesh] OR China[Title/Abstract] OR Chinese[Title/Abstract])  
AND (seasonality[Title/Abstract] OR seasonal[Title/Abstract]  
OR "time series"[Title/Abstract] OR temporal[Title/Abstract])
```

For example, the query for Xi'an was:

```
("Influenza, Human"[Mesh] OR influenza[Title/Abstract] OR ILI[Title/Abstract])  
AND (Xi'an[Title/Abstract] OR Xian[Title/Abstract])  
AND (China[Mesh] OR China[Title/Abstract] OR Chinese[Title/Abstract])  
AND (seasonality[Title/Abstract] OR seasonal[Title/Abstract]  
OR "time series"[Title/Abstract] OR temporal[Title/Abstract])
```

Eligibility criteria... Influenza studies were considered eligible if they reported prefecture-/city-level surveillance (administrative level two) with at least five consecutive years of observations (≥ 60 months), monthly or weekly temporal resolution, and an influenza indicator suitable for time-series modelling (e.g., laboratory-confirmed influenza positivity among ILI/ARI consultations or sentinel specimens). Studies were excluded if the full text could not be obtained, if reported data were incomplete (e.g., interrupted series without sufficient values to reconstruct a continuous time series), or if the data source and/or surveillance definition was unclear or could not be verified from the article.

To ensure suitability for modelling both within-year seasonality and between-year variation, all candidate studies were manually checked to confirm that extracted series were continuous and year-indexed. Studies were excluded if they reported only an aggregated seasonal profile (e.g., multi-year mean/median for each calendar month, resulting in 12 values with the x-axis labelled Jan–Dec, or analogous week-of-year summaries) rather than a multi-year time series with a year-month/year-week index.

Selection of one influenza series per city... When multiple eligible influenza series were available for a given city, one dataset was selected based on a predefined priority order: (1) continuous time-series length of ≥ 5 years (required); (2) temporal resolution (weekly preferred over monthly, if comparable); (3) longest uninterrupted time coverage. Where ties remained, we preferred the series with the most specific virological definition (e.g., laboratory-confirmed influenza A/B rather than syndromic outcomes alone).

The final dataset included one influenza time series for each of the eight cities. Data sources and surveillance indicator definitions for each city–pathogen pair are provided in Supplementary Table S1.

Table 5: Supplementary Table S1. Surveillance indicator definitions for RSV and influenza (InfV) across included Chinese cities

City	Pathogen	Indicator definition	Clinical setting	Laboratory method	Reference
Beijing	RSV	RSV antigen positivity in ARI inpatients	Inpatient (ARI)	IFA/DFA on NPA	[a]
Beijing	InfV	Influenza-positive samples by subtype	Outpatient (ILI sentinel)	Virus isolation	[b]
Xi'an	RSV	RSV-positive cases / positivity rate	Out-/inpatient (ARI)	Not reported	[c]
Xi'an	InfV	Daily ILI case counts (syndromic)	Outpatient (ILI)	None	[d]
Lanzhou	RSV	RSV positivity; RSV-A genotypes	Inpatient (ARI)	RT-PCR + subgroup PCR + sequencing	[e]
Lanzhou	InfV	Laboratory-confirmed influenza	Out-/inpatient	RT-PCR / isolation / serology	[f]
Wuhan	RSV	RSV positivity in ARTI inpatients	Inpatient (ARTI)	DFA on NP swabs	[g]
Wuhan	InfV	ILI% and influenza-positive specimens	Outpatient (ILI sentinel)	RT-PCR	[h]
Wenzhou	RSV	RSV-positive LRTI inpatients	Inpatient (LRTI)	DFA	[i]
Wenzhou	InfV	Influenza-positive LRTI inpatients	Inpatient (LRTI)	DFA / culture / ELISA	[j]
Suzhou	RSV	RSV positivity in neonatal LRTI inpatients	Inpatient (neonatal LRTI)	DFA on NPA; PCR for atypical agents	[k]
Suzhou	InfV	Influenza positivity in SARI inpatients	Inpatient (SARI)	RT-PCR on NPA	[l]

References

- [1] Diamond, C., Gong, H., Sun, F.Y., Liu, Y., Quilty, B.J., Jit, M., Yang, J., Yu, H., Edmunds, W.J., Baguelin, M., 2022. Regional-based within-year seasonal variations in influenza-related health outcomes across mainland china: a systematic review and spatio-temporal analysis. *BMC Medicine* 20, 58. doi:[10.1186/s12916-022-02269-5](https://doi.org/10.1186/s12916-022-02269-5).
- [2] Guo, L., Deng, S., Sun, S., Wang, X., Li, Y., 2024. Respiratory syncytial virus seasonality, transmission zones, and implications for seasonal prevention strategy in china: a systematic analysis. *The Lancet Global Health* 12, e1005–e1016.
- [3] Li, J., Chen, Y., Wang, X., Yu, H., 2021. Influenza-associated disease burden in mainland china: a systematic review and meta-analysis. *Scientific Reports* 11, 2886. doi:[10.1038/s41598-021-82161-z](https://doi.org/10.1038/s41598-021-82161-z).